Saad Alif A Zaman

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Research Interests

- **Data-efficient deep learning for visual inspection:** instance segmentation under limited labels
- **Vision-based Structural Health Monitoring (SHM) and Non-Destructive Evaluation (NDE):** condition assessment of steel, marine, and civil infrastructure (corrosion, biofouling)
- **Toward autonomous inspection:** perception systems for UAV, ROV, and robotic platforms

Education

B.Sc. In Naval Architecture And Marine Engineering (NAME) (April 2019 – July 2024)
Bangladesh University of Engineering and Technology (BUET)
Cumulative GPA: 3.37 / 4.00

Relevant Coursework:
- **Computation & Mathematics:** Computer Programming I & II (C/C++, FORTRAN, OOP) [A+]; Statistics, PDEs, and Matrices (Applied Probability & Linear Algebra) [A]; Numerical Computations [B]
- **Structural Health & Degradation:** Marine Maintenance and Repair (Corrosion, Fatigue, Biofouling & Defect Analysis) [A-]; Welding Technology (NDT & Defect Inspection) [B+]; Finite Element Method for Ship Structure [B+]
- **Mechanics & Materials:** Mechanics of Structure [A]; Shipbuilding Materials [B+]

Research Experience

➢ **Independent Research Project** (August 2025 – Present)
"Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection"
Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept. of Naval Architecture and Marine Engineering, BUET
Samiul Based Shuvo, Assistant Professor, Dept. of Biomedical Engineering, BUET

- **Proposed** a k-n Fold Augmentation Cross-Validation (k-n Fold ACV) protocol and a novel Interaction Effect (IE) Metric to quantify order sensitivity among data augmentations under extreme data scarcity.
- **Validated** the framework on a newly curated NDBD dataset, comprising 35 high-resolution ship hull images with 92 annotated biofouling instances.
- **Analyzed** 110 pairwise combinations of 11 domain-specific augmentations using YOLOv8m-seg as the base architecture.
- **Executed** cross-validation across YOLO11m-seg and Mask R-CNN, and made the code and dataset publicly available to ensure reproducibility.

➢ **Undergraduate Thesis** Final Grade: A+ (4.00/4.00) (June 2023 – July 2024)
"Developing a biofouling & corrosion identification and distribution system using video images based on deep learning"
Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept of Naval Architecture and Marine Engineering, BUET

- **Collected and curated** the Narayanganj Dockyard Biofouling Dataset (NDBD) directly from Dockyard & Engineering Works Ltd., Narayanganj (CC BY-NC 4.0, publicly available on GitHub), supplementing it with public bridge inspection data for steel corrosion severity.
- **Trained** YOLOv8 segmentation variants via transfer learning to simultaneously detect marine biofouling and structural corrosion, achieving a peak fouling F1 score of 0.90.
- **Developed** a video-based detection prototype for automated hull inspection.
- **Secured** the competitive RISE Student Research Grant to fund the system's development.

➢ **Junior Year Project** Final Grade: A+ (4.00/4.00) (May 2022 – May 2023)
"Designing an Inland 100 Passenger Ship for Inland waterways of Bangladesh"
Supervisor: Dr. Md. Mashiur Rahaman, Professor, Dept of Naval Architecture and Marine Engineering, BUET

- **Designed** a conceptual 100-passenger inland vessel integrating a tailored electric propulsion system.
- **Optimized** the design specifically for Bangladesh's inland waterways to reduce CO_2 emissions and significantly improve energy efficiency over conventional diesel systems.

Publications

1. Minhazul Islam, **Saad Alif A Zaman**, Md. Mashiur Rahaman, "Concept of an Electric-Powered Passenger Vessel in Bangladesh," Journal of Maritime Research, vol. 20, no. 3, pp. 68–71, Dec. 2023
2. **Saad Alif A Zaman**, Samiul Based Shuvo, Kazi Naimul Hoque, "Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection," Preprint; in revision following peer review, targeting resubmission, 2026. [SSRN link]

RESEARCH GRANTS, RECOGNITION & SCHOLARSHIPS

- **RISE Student Research Grant 2023:**
  Awarded a competitive Student Research Grant (BDT 65,790) by **R**esearch and **I**nnovation center for **S**cience and **E**ngineering (RISE) for the research project: "*Developing a biofouling & corrosion identification and distribution system using video images based on deep learning.*"
- **SPARRSO Research Fellowship Program 2025–26:**
  Shortlisted (final round) among national applicants for a proposal-stage research fellowship with the Space Research and Remote Sensing Organization (SPARRSO), Bangladesh; invited to present the written proposal "*Development of a Fixed-Wing VTOL Quad Plane for Flood Monitoring & Disaster Response*" (design not yet implemented).
- **University Stipend Scholarship-2019, 2020, 2021 – BUET:**
  Competitive merit-based financial award granted to high-performing undergraduate students based on academic results.
- **Rajshahi Board Scholarship HSC 2018:**
  Merit-based national scholarship awarded for outstanding academic performance in the Higher Secondary Certificate (Grade 12 equivalent) examination.

STANDARDIZED TEST SCORES

International English Language Testing System (IELTS) Academic (15th December, 2025)
Overall Band Score: 7.5 (Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 7.0)

TEACHING & ACADEMIC EXPERIENCE

Materials Development Instructor
Udvash Academic & Admission Care | Dhaka, Bangladesh (April 2025 – August 2025)
● **Evaluated** question quality, solution accuracy, and curriculum alignment for high-stakes engineering and university admission examinations.
● **Developed and reviewed** rigorous assessment materials and model tests, demonstrating the meticulous grading and content-management skills required for university-level course support.

Academic Mentor (Mathematics & Physics) (2019 – Present)
● **Instruct** secondary and higher-secondary students in foundational and advanced mathematics and physics through targeted small-batch and one-to-one sessions.
● **Guide** students through complex problem-solving methodologies across both English and Bangla medium curricula, leading to multiple successful admissions to top-tier engineering institutions including BUET, KUET, DU, and IUT.

Industrial Internship – Shipyard Practice
Khulna Shipyard Limited | Khulna, Bangladesh (October 2022 – November 2022)
● **Completed** industrial shipyard practice, gaining direct exposure to the physical assets, maintenance environments, and operational constraints of maritime infrastructure.

TECHNICAL SKILLS

- **Programming:** Python, C, C++
- **Deep Learning**: Ultralytics YOLO, Mask R-CNN, PyTorch, TensorFlow
- **Data Annotation & Processing**: CVAT, Roboflow, LabelImg
- **Research Computing & Collaboration**: Google Colab, Kaggle
- **CAD software**: AutoCAD, Rhino
- **Finite Element Analysis software:** Abaqus
- **Document Preparation**: LaTeX, Prism, Microsoft Office

LEADERSHIP & VOLUNTEERING

● **Founder, Monthly Reading Initiative (Telegram):** (March 2024 – Present)
Founded a monthly book and magazine reading platform on Telegram, which continues to be maintained by junior students and now includes regular quizzes and discussion activities.
● **Badhon BUET Zone, Volunteer Member:** (May 2019 – July, 2024, Dhaka)
Active volunteer in a student-led blood donation organization; donated blood multiple times and participated in donation drives and awareness activities.

REFERENCES

Dr. Kazi Naimul Hoque
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and Research Advisor
Dept. of Naval Architecture and Marine Engineering,
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Faculty Profile

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Assistant Professor, Research Advisor
Dept. of Biomedical Engineering
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Faculty Profile

Dr. Md. Mashud Karim
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